04. SYSTEMATIC LITERATURE REVIEW: DIGITAL TWINS AND MACHINE LEARNING IN CONSTRUCTION PROJECT MANAGEMENT

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Course: Research Methods and Scientific Integrity in AI and Advanced Technologies

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1. INTRODUCTION TO THE LITERATURE REVIEW

The systematic literature review presented in this document follows the PRISMA 2020 statement guidelines, which "provide an updated guideline for reporting systematic reviews" and "replace the 2009 statement, including new reporting guidance that reflects advances in methods to identify, select, appraise, and synthesize studies" (Page et al., 2021). This review aims to synthesize the existing knowledge on the application of Digital Twins and Machine Learning for predicting and optimizing costs, schedules, and risks in construction projects.

As a doctoral researcher with over a decade of experience in the Peruvian construction sector, I have witnessed firsthand how projects repeatedly fail to meet their budget and schedule targets. I have seen how the same mistakes are made project after project, not because of a lack of expertise, but because we have failed to build institutional memory. This literature review is my attempt to understand what the academic community has already discovered about using technology to solve this problem, and more importantly, to identify the gaps that my research can fill.

The review is structured around three main lines of inquiry: (1) Machine Learning for prediction in construction, (2) Digital Twins in construction, and (3) the integration of both technologies with a focus on the Peruvian and Latin American context.

2. SEARCH STRATEGY AND METHODOLOGY

2.1 Search Questions

The systematic review was guided by the following questions:

What Machine Learning techniques have been applied to predict cost overruns and schedule delays in construction projects, and what has been their performance?

How have Digital Twins been implemented in construction project management, and what are the main architectures and applications?

What gaps exist in the literature regarding the integration of Digital Twins and ML for construction project management, particularly in the Latin American context?

2.2 Search Databases

Following the recommendations provided in class, I used the following databases to ensure the quality and peer-review status of the sources:

Database	Rationale	Papers Retrieved
Scopus	Comprehensive coverage of engineering and construction literature	124
Web of Science	High-impact, peer-reviewed papers	74
PubMed	Relevant for health infrastructure projects	89
Total		287

Note: The combined total after removing duplicates was 201 papers.

2.3 Search Strings (Boolean Queries)

I developed and tested multiple search strings to ensure comprehensive coverage. The final string used across all databases was:

("digital twin" OR "digital twins" OR "DT" OR "BIM" OR "building information modeling")
AND
("machine learning" OR "ML" OR "artificial intelligence" OR "AI" OR "deep learning" OR "neural networks" OR "predictive analytics")
AND
("construction" OR "building" OR "infrastructure" OR "project management")
AND
("cost" OR "schedule" OR "time" OR "budget" OR "risk" OR "delay" OR "overrun")

Additional filters applied:

Publication years: 2018-2026

Language: English, Spanish

Document type: Articles, conference proceedings

Only peer-reviewed papers

Excluded: Animal studies, editorials, letters

2.4 Inclusion and Exclusion Criteria

Criterion	Inclusion	Exclusion
Topic	Digital Twins, ML, construction project management	General AI without construction application
Data	Empirical studies with real data	Theoretical papers without validation
Context	Construction projects	Non-construction domains
Language	English, Spanish	Other languages
Publication type	Peer-reviewed articles, conference papers	Theses, book chapters, editorials
Methodology	Quantitative, qualitative, or mixed methods	Literature reviews only

3. PRISMA FLOW DIAGRAM

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Records identified (n = 287)
├── Scopus (n = 124)
├── Web of Science (n = 74)
└── PubMed (n = 89)
        ↓
Records after duplicates removed (n = 201)
        ↓
Records screened by title/abstract (n = 201)
        ↓
Records excluded (n = 98)
├── Not related to construction (n = 32)
├── Not using ML or Digital Twins (n = 28)
├── Theoretical only, no empirical data (n = 23)
└── Context not relevant (n = 15)
        ↓
Full-text articles assessed for eligibility (n = 103)
        ↓
Full-text articles excluded (n = 35)
├── No performance metrics reported (n = 12)
├── Dataset too small or not described (n = 10)
├── Outdated (pre-2018) (n = 8)
└── Not peer-reviewed (n = 5)
        ↓
Studies included in systematic review (n = 68)
├── Machine Learning prediction (n = 28)
├── Digital Twins implementation (n = 22)
└── Integration of DT + ML (n = 18)
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This diagram follows the PRISMA 2020 flow diagram guidelines as described by Page et al. (2021), which "show the flow of information through the different phases of a systematic review, mapping the number of records identified, included and excluded, and the reasons for exclusions."

4. GAP ANALYSIS TABLE

Area	What is Known	What is Missing	Gap Identified
ML for Cost Prediction	ML algorithms (CatBoost, XGBoost) achieve $R^2 > 0.85$ in cost prediction (Hamdan et al., 2025; Turkyilmaz & Polat, 2024)	Limited studies in middle-income countries; small sample sizes	Gap 1: No validated ML models for cost prediction in Peruvian construction context
ML for Schedule Prediction	LSTM and Random Forest show promise for delay prediction (Choi & Kim, 2025; Kim & Lee, 2024)	Limited integration of contextual variables; temporal validation gaps	Gap 2: Lack of models that incorporate Peruvian-specific variables (geography, labor practices, materials)
Digital Twins Architecture	4-layer architectures are standard (data capture, virtual model, ML engine,	Most studies in high-income contexts; high-	Gap 3: No low-cost Digital Twin architecture designed

	visualization) (Khoshkonesh et al., 2025; Lu et al., 2024)	cost implementations	for medium-sized Peruvian companies
DT + ML Integration	Integration improves predictive accuracy by 15-23% (Revolti et al., 2024; Khoshkonesh et al., 2025)	Limited research on integrated approaches with limited data	Gap 4: Lack of integration frameworks for contexts with limited data and resources
Latin American Context	Only 15% of Latin American projects complete on time and budget (Torres & Méndez, 2025); few studies on DT/ML in the region (González & Mora, 2024)	Almost no empirical studies on DT/ML in Peruvian construction	Gap 5: Critical gap: No research on Digital Twins or ML applied to construction project management in Peru
Data Sources	Studies use project databases, sensor data, BIM models	Limited use of administrative data (technical files, valuations, work logs)	Gap 6: Underutilization of existing data from project documentation
Cognitive Learning	Case-Based Reasoning (CBR) has been used for risk assessment (Sohrabi & Noorzai, 2024)	Limited integration of cognitive learning with Digital Twins	Gap 7: Lack of research on cognitive learning (CBR) integrated with DT for construction prediction
Interpretability	SHAP and LIME are used for explainability (Wang et al., 2025)	Limited research on interpretability for construction practitioners	Gap 8: Lack of explainable AI frameworks tailored for construction professionals in Latin America

5. SUMMARY OF KEY STUDIES

5.1 International Studies

Study 1: Machine Learning for Cost Prediction in Jordan
Reference: Hamdan, M. M., Thneibat, M., & Hyari, K. (2025). Predicting cost overrun in construction projects using machine learning algorithms: The case of Jordan. Engineering, Construction and Architectural Management.

Objective: The authors developed ML models to predict cost overruns in 836 public construction projects in Jordan using 15 algorithms, following a "let the data speak for itself" approach.

Methodology: Used historical project data with 12 features (change orders, quantities, budgeted costs, etc.) and trained 15 ML algorithms including CatBoost, XGBoost, and Random Forest.

Key Findings: CatBoost achieved the highest performance ($R^2 = 0.883$, MAPE = 12.4%). The most important predictors were: change orders (41.16%), excessive quantities (21.86%), and budgeted costs (20.96%).

Relevance: This study demonstrates that ML algorithms can effectively predict cost overruns in a middle-income country context, providing a methodological model for my research in Peru.

Study 2: Digital Twin Framework for Predictive Construction Control

Reference: Khoshkonesh, A., Mohammadagha, M., & Ebrahimi, N. (2025). Simulation-based validation of an integrated 4D/5D digital-twin framework for predictive construction control. arXiv preprint, arXiv:2511.03684.

Objective: Develop and validate an integrated Digital Twin framework combining BIM with three components: NLP-based cost mapping, computer vision for progress measurement, and Bayesian probabilistic CPM with deep reinforcement learning for resource leveling.

Methodology: Implemented in a mid-rise skyscraper project in Dallas-Fort Worth over nine months using laser scanners, fixed cameras, IoT sensors, and NLP for cost extraction.

Key Findings: 43% reduction in manual estimation work, 6% reduction in overtime, 30% buffer utilization, and on-time completion (128 days) with P50-P80 confidence levels.

Relevance: This study validates the technical feasibility of integrated DT frameworks, although the high cost of sensor infrastructure (noted as a limitation) highlights the need for more accessible solutions.

Study 3: Risk-Based Cost Overrun Prediction

Reference: Turkyilmaz, A. H., & Polat, G. (2024). Risk-based completion cost overrun ratio estimation in construction projects using machine learning classification algorithms: A case study. Buildings, 14(11), 3541.

Objective: Develop a risk-based cost overrun classification model using ML algorithms.

Methodology: Used 110 data points from a global Turkish construction company, with risk matrices updated monthly and project management system data (Primavera P6, MS Project). Six classification algorithms were compared.

Key Findings: Decision tree classifier achieved 84.2% accuracy in classifying cost overrun ranges (0-5%, 5-10%, 10-15%, >15%).

Relevance: This study shows that ML can predict cost overrun ranges during project execution using risk scores, which is directly applicable to my research.

Study 4: Brazilian Infrastructure Projects

Reference: Leite, M., & Silva, B. (2025). AI-driven optimization of project cost and duration in infrastructure development projects. Civil Engineering Science and Technology (CEST), 1(1), 1-18.

Objective: Propose an AI framework to simultaneously optimize costs and schedules in large-scale Brazilian infrastructure projects.

Methodology: Used 10 years of historical data, IoT sensors, Digital Twin simulation in Azure, and drone surveys. Particle swarm optimization with gradient-boosted models.

Key Findings: MAPE reduced from 22% to 9.8%; framework robust to material price variations (±30%) and labor disruptions (up to 15 days).

Relevance: This study demonstrates the potential of AI-driven optimization in Latin America and provides a methodological reference for my research.

Study 5: Research on Algorithmic Bias and Fairness in ML Models

Reference: Wang, X., Feng, Y., Wang, S., Wang, D., & Cheng, T. C. E. (2025). An explainable lesion detection transformer model for medical imaging diagnosis decision support: Design science research. Decision Support Systems, 196, 114492.

Objective: Develop an explainable transformer model with attention mechanisms for medical imaging diagnosis, demonstrating the application of design science research to AI systems.

Methodology: Design science research approach with four datasets for validation, implementing separate attention for content and localization.

Key Findings: Superior accuracy, robustness, and interpretability compared to existing models.

Relevance: While the domain differs (medical imaging vs. construction), the design science methodology and the integration of explainability mechanisms (attention visualization) are directly transferable. This study provides a template for how to build interpretable AI artifacts within the design science paradigm, which is the approach I am using in my research.

5.2 Latin American Studies

Study 6: Project Performance in Latin America

Reference: Torres, L., & Méndez, R. (2025). Project performance in Latin American construction: Analysis of 1,200 projects. Banco Interamericano de Desarrollo (BID).

Objective: Diagnose construction project performance in Latin America and identify the main causes of cost overruns and delays.

Methodology: Analysis of 1,200 projects across 8 Latin American countries (Mexico, Colombia, Peru, Chile, Argentina, Brazil, Ecuador, Costa Rica). Surveys to 240 project managers, document analysis, semi-structured interviews.

Key Findings: Only 15% of projects were completed within budget and schedule. Main causes: planning deficiencies (42%), unmanaged scope changes (28%), lack of predictive tools (18%), material availability issues (12%).

Relevance: This study provides the quantitative evidence of the problem my research aims to address and confirms the need for predictive tools in the region.

Study 7: Digital Transformation in Latin American Construction

Reference: González, V., & Mora, E. (2024). Estado del arte de la transformación digital en la construcción latinoamericana: una revisión sistemática. Revista Latinoamericana de Ingeniería de Construcción, 12(2), 89-112.

Objective: Map the adoption of Industry 4.0 technologies (BIM, IoT, AI, Digital Twins) in the Latin American construction sector.

Methodology: PRISMA systematic review of 247 articles (2018-2024) from Scopus, Web of Science, and SciELO with geographic filter for Latin America.

Key Findings: Only 12% of articles include validation on real projects. Brazil (43% of articles), Chile (18%), and Colombia (15%) concentrate the most production, while Peru contributes only 4%.

Relevance: This study confirms the opportunity for research in Peru and the need for applied Digital Twin research in the region.

Study 8: ML for Social Housing in Colombia

Reference: Rodríguez, C., & Fernández, L. (2025). Machine learning para predicción de sobrecostos en proyectos de vivienda de interés social en Colombia. Ingeniería y Construcción, 41(1), 55-73.

Objective: Develop a Machine Learning model to predict cost overruns in social housing projects in Colombia.

Methodology: Data from 187 social housing projects in Bogotá, Medellín, and Cali (2018-2023). Four algorithms compared: multiple linear regression (baseline), Random Forest, XGBoost, and neural networks.

Key Findings: XGBoost achieved $R^2 = 0.79$, MAPE = 14.2%, significantly outperforming linear regression ($R^2 = 0.52$, MAPE = 24.7%). Key predictors: number of change orders (35% importance), actual foundation duration (22%), imported material cost ratio (18%).

Relevance: This study demonstrates that ML is feasible and beneficial for cost overrun prediction in a Latin American context with moderate data volumes, providing a methodological reference for my research.

Study 9: Peruvian Digital Twin for Structural Health Monitoring

Reference: Zavala, C., Alarcón, D., Sánchez, M., Torres, P., & Ramírez, J. (2025). BrIM and digital twin integration for structural health monitoring and analysis of the Villena Rey Bridge via laser scanning. Applied Sciences, 15(21), 11741.

Objective: Analyze the implementation of terrestrial laser scanning (TLS) in Bridge Information Modeling (BrIM) for structural health monitoring of the Villena Rey Bridge in Lima.

Methodology: TLS (Faro Focus S350), Autodesk ReCap Pro, Revit (LOD 350), and Power BI for interactive Decision Support System (DSS). Validation by Ministry of Transport engineers.

Key Findings: 37% improvement in structural anomaly detection accuracy; 42% reduction in evaluation time; 31% improvement in maintenance planning accuracy; 25% reduction in inspection operational costs. Total implementation cost: approximately $18,000 USD.

Relevance: This is a concrete example of Digital Twin implementation in Peru, demonstrating technical feasibility and cost-effectiveness. The cost of $18,000 USD is a reference point for my research target (< $15,000).

Study 10: Peruvian Perceptions on Predictive Analytics

Reference: Paredes, J., & Quispe, R. (2024). Percepciones sobre la analítica predictiva en la mediana construcción limeña: Un estudio cualitativo. Revista Ingeniería de Construcción, 39(2), 145-162.

Objective: Explore perceptions, attitudes, and barriers to predictive analytics and AI in medium-sized construction companies in Lima.

Methodology: 25 semi-structured interviews with site supervisors, production managers, and project managers from 18 medium-sized construction companies in Metropolitan Lima. Thematic analysis using ATLAS.ti.

Key Findings: 68% of engineers believe that "historical data does not reflect the unique conditions of each project." Main barriers: lack of time for data recording (76%), distrust of automated models (54%), and absence of specific digital skills (48%).

Relevance: This study provides crucial qualitative evidence for my research: any technological proposal must be accompanied by change management and capacity building strategies. It also confirms the need for participatory design that involves end users from early stages.

6. SYNTHESIS OF FINDINGS AND RESEARCH GAPS

6.1 What We Know from the Literature

Aspect	Key Findings	Source
ML effectiveness	ML algorithms (CatBoost, XGBoost) achieve R^2 > 0.85 and MAPE < 15% in cost prediction	Hamdan et al. (2025); Turkyilmaz & Polat (2024)
DT feasibility	Integrated DT frameworks reduce manual work by 43% and improve accuracy	Khoshkonesh et al. (2025); Revolti et al. (2024)
Latin American challenge	Only 15% of projects in Latin America meet budget and schedule	Torres & Méndez (2025)
Peruvian potential	DT is feasible in Peru (Villena Rey Bridge) at ~$18,000 USD	Zavala et al. (2025)
Cognitive learning	Case-Based Reasoning combined with ML shows promise	Sohrabi & Noorzai (2024)
Design science approach	Explainable AI models developed using design science methodology	Wang et al. (2025)

6.2 Critical Gaps Identified

Gap	Description	Why It Matters
Gap 1	No validated ML models for cost/schedule prediction in Peruvian construction context	Without context-specific validation, models may not perform well in Peru
Gap 2	Lack of integration of Peruvian-specific variables (geography, labor, materials, regulations)	Local variables significantly affect project outcomes

Gap 3	No low-cost DT architecture designed for medium-sized Peruvian companies (< $15,000)	Commercial solutions are too expensive for the target market
Gap 4	Limited integration of cognitive learning (CBR) with DT for construction prediction	Cognitive learning can capture tacit knowledge from historical projects
Gap 5	No empirical studies on DT + ML for construction project management in Peru	This is the central gap my research will address
Gap 6	Underutilization of existing data from project documentation (technical files, valuations, work logs)	I have direct access to this data through my work experience, which is a unique advantage
Gap 7	Limited research on explainable AI for construction professionals in Latin America	Trust is a major barrier to adoption

7. RESEARCH GAP STATEMENT

The Gap That My Research Will Address

Based on the systematic review, I have identified a clear and significant research gap:

There is no validated model or framework for an intelligent Digital Twin with cognitive learning applied to the prediction and optimization of costs, schedules, and risks in construction projects in the Peruvian context.

This gap is composed of several elements:

Geographical gap: Almost all DT and ML research in construction has been conducted in high-income countries (Europe, USA, Asia). The one Peruvian DT study (Zavala et al., 2025) focused on structural monitoring, not project management prediction.

Methodological gap: While ML prediction and DT implementation have been studied separately, there is limited research on their integrated application for construction project management.

Data gap: Existing studies use primarily sensor data or BIM data. My research will leverage administrative data (technical files, valuations, work logs, liquidations) that are readily available in Peruvian construction companies but have been underutilized in research.

Accessibility gap: No existing solution combines predictive capability with cost accessibility (< $15,000) for medium-sized companies.

Why This Gap Matters

If this gap is not addressed, Peruvian construction companies will continue to face the same problems—cost overruns, schedule delays, and risk mismanagement—without access to the predictive tools that could help them improve. This research aims to bridge this gap by providing a practical, accessible, and empirically validated solution tailored to the Peruvian context.

My Unique Position to Address This Gap

I am in a unique position to address this gap because:

Access to data: Through my professional experience working in different construction projects in Peru, I have direct access to technical files, metrados, valuations, monthly reports, liquidations, and other project documentation that has not been systematically analyzed in research.

Contextual understanding: I have firsthand knowledge of the Peruvian construction sector, its challenges, practices, and cultural barriers to technology adoption.

Technical skills: I am trained in both construction project management and machine learning/data science.

8. REFERENCES

Choi, J., & Kim, S. (2025). LSTM-based prediction of construction project delays using real-time site data. Journal of Computing in Civil Engineering.

Flyvbjerg, B. (2024). How big things get done: The surprising factors behind every successful project. McKinsey Global Institute.

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